An angular Lyapunov correction for a partially damped map at a fourth-order resonance

Abstract

We construct a local strict Lyapunov function for one four-dimensional partially damped recurrence whose linearization has the multipliers $i, -i$ and a strictly stable complementary pair. A positive quartic leading function has an indefinite eighth-degree discrete drift, and every homogeneous polynomial sextic correction in a specified family still has a sign-changing leading defect. A periodic angular correction, homogeneous of degree six and $C^{5,1}$ at the origin, gives a negative fourth-step decrement. We retain the nonzero finite-graph residual, control the transverse phase and convert this decrement into a one-step state function. An invariant sublevel neighborhood then gives local asymptotic stability and componentwise upper decay rates for independently initialized original phases. The result concerns one fixed map and step; the averaging and finite-step Lyapunov devices are classical.

Keywords: Lyapunov function; discrete dynamical system; partial damping; resonance; local asymptotic stability

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1 The map and the energy

At a root-of-unity resonance, a scalar nonresonant radial coefficient need not determine the behavior of the actual iterates. For multipliers $i, -i$, the conjugate-cubic term must be retained along with the usual cubic term; see Kuznetsov [1, Section 9.5.5]. A leading invariant can organize the calculation without providing a strict energy. The question considered here is whether the next discrete loss can be turned into a positive state function that controls the full original phase.

Set $q = \sqrt{2}$. For $p = (x, v, y, w) \in \mathbb{R}^4$, define $T(p) = (x^+, v^+, y^+, w^+)$ by

$$\begin{aligned} v^+ &= \frac{v - q(x + y^2)}{1 + q}, & x^+ &= x + qv^+, \\ w^+ &= w - q(y + 2xy + 4y^3), & y^+ &= y + qw^+. \end{aligned} \tag{1}$$

This is the fixed step $h = \sqrt{2}$ semi-implicit update for the potential

$$F(x, y) = \frac{x^2 + y^2}{2} + xy^2 + y^4 = \frac{(x + y^2)^2}{2} + \frac{y^2 + y^4}{2}, \tag{2}$$

with unit damping only in the v coordinate. All coefficients in (1) are fixed. The polynomial F is coercive and has its unique critical minimum at the origin. We do not assume that it is globally convex, or that mechanical energy is decreasing for this step.

Write $z = (X, Y)$ and $u = (x, v)$, and introduce

$$X = \frac{y}{2}, \qquad Y = \frac{y}{2} - \frac{w}{q}, \qquad S = \begin{pmatrix} 3-2q & 2-q \\ q-2 & q-1 \end{pmatrix}, \qquad b = \begin{pmatrix} 8-8q \\ -8+4q \end{pmatrix}. \tag{3}$$

The following finite polynomial graph is part of the energy definition:

$$\begin{aligned} H_2(z) &= \big((2q-4)X^2 - 2qY^2,\ (4-2q)(X^2 - Y^2)\big), \\ H_4(z) &= \big((32q-64)XY(X^2+Y^2), \\ &\qquad\qquad (64-96q)X^3Y + (128-32q)XY^3\big), \\ H = H_2 + H_4, &\qquad \sigma = u - H(z), \qquad \Xi(p) = (z, \sigma). \end{aligned} \tag{4}$$

It is not an invariant graph. In these coordinates the exact map $\Psi = \Xi T \Xi^{-1}$ is

$$\begin{aligned} u &= \sigma + H(z), \qquad z^+ = (-Y - 4u_1 X - 32X^3, X), \\ \sigma^+ &= Su + bX^2 - H(z^+). \end{aligned} \tag{5}$$

Let $(z_j(z,\sigma), \sigma_j(z,\sigma)) = \Psi^j(z,\sigma)$. Thus a subscript denotes an actual finite iterate, with no graph restriction imposed after the initial state.

Define the quartic and sextic polynomials

$$\begin{aligned} Q(z) &= X^4 + Y^4 - 2(q-1)X^2Y^2, \\ A(z) &= -16q(X^5Y + XY^5) + 64X^3Y^3. \end{aligned} \tag{6}$$

For an angular variable θ, put

$$\begin{aligned} t(\theta) = \sin^2(2\theta), &\qquad e(\theta) = 1 - \frac{q}{2}t(\theta), \\ f(\theta) = t(\theta)\left(q - 2 + \frac{q+1}{4}t(\theta)\right), &\qquad \langle g \rangle = \frac{1}{2\pi}\int_0^{2\pi} g(\theta)\,\mathrm{d}\theta, \\ I_0 = \langle e^{-1/2} \rangle, &\qquad J_0 = \langle f e^{-5/2} \rangle, \qquad c = \frac{1024 J_0}{I_0}, \\ \eta(\theta) = \frac{1}{16(2+q)}\int_0^\theta &\left(c e(s)^{-1/2} - 1024 f(s) e(s)^{-5/2}\right)\,\mathrm{d}s. \end{aligned} \tag{7}$$

The correction and the center energy are

$$B(z) = Q(z)^{3/2}\eta(\arg(X + iY)) \quad (z \neq 0), \qquad B(0) = 0, \qquad U = Q + A + B. \tag{8}$$

The function η is periodic, so the argument branch in this formula is immaterial. We will prove that $J_0 < -1/25$ and $c < -512/25$.

To define the transverse weight, set

$$P = \frac{1}{136}\begin{pmatrix} 123 + 12q & 96 - 69q \\ 96 - 69q & 247 - 75q \end{pmatrix}, \qquad \kappa = -\frac{c}{(2+q)^2}. \tag{9}$$

Use Euclidean norms and define the single fixed constant

$$C_* = \sup_{\substack{0<\|z\|\le 1\\ 0<\|\sigma\|\le 1}} \frac{|U(z_4(z,\sigma)) - U(z_4(z,0))|}{\|z\|^4\|\sigma\|}, \qquad K = 1 + \frac{8C_*^2}{\kappa}. \tag{10}$$

Finiteness of this supremum is proved on its entire stated domain in Lemma 4, before it is used to choose a local radius. In particular, it does not depend on an unknown bound for an infinite orbit. Finally, put

$$W_0(z,\sigma) = U(z) + K\sigma^T P\sigma, \qquad \Phi = \sum_{j=0}^{3} W_0 \circ \Psi^j, \qquad V = \Phi \circ \Xi. \tag{11}$$

This is a fixed state function evaluated through four known finite iterates.

Theorem 1. *For the map* (1)*, the definitions* (7)–(11) *are well posed. The function* B *is* $C^{5,1}$ *at zero,* P *is positive definite, and* V *is* C^5*. There are constants* $r_0, m, M, a > 0$ *such that, with* $r = \|z\|$ *and* $s = \|\sigma\|$*,*

$$\begin{aligned} m(r^4 + s^2) \le \Phi(z,\sigma) \le M(r^4 + s^2),\\ \Phi(\Psi(z,\sigma)) - \Phi(z,\sigma) \le -a(r^8 + s^2) \qquad \big(\|(z,\sigma)\| < r_0\big). \end{aligned} \tag{12}$$

There is an open neighborhood $\mathcal{B}$ *of the original origin, invariant under* T*, on which every original orbit converges to zero. The origin is Lyapunov stable in the full original phase. For one* $k > 0$ *common to* $\mathcal{B}$*,*

$$V(T^n p_0) \le \frac{V(p_0)}{1 + knV(p_0)} \qquad (p_0 \in \mathcal{B},\ n \ge 0). \tag{13}$$

For each such initial phase,

$$\|(y_n, w_n)\| = O(n^{-1/4}), \qquad \|(x_n, v_n)\| = O(n^{-1/2}). \tag{14}$$

The initial positions and velocities in $\mathcal{B}$ *are independent; no graph initialization is required.*

There is also a precise limitation on polynomial corrections to the leading function Q.

Theorem 2. *For every real homogeneous polynomial* Q_6 *of degree six, the limit*

$$D_{Q_6}(z) = \lim_{\varepsilon\to 0} \varepsilon^{-8}\left[(Q + Q_6)(z_4(\varepsilon z, 0)) - (Q + Q_6)(\varepsilon z)\right] \tag{15}$$

is a homogeneous polynomial of degree eight which takes both strictly positive and strictly negative values. The assertion concerns this leading quartic, these coordinates and this sextic polynomial family.

Theorem 2 does not exclude other leading energies or nonpolynomial corrections. Theorems 1 and 2 together exhibit the role of the angular term in this particular construction.

The rates in (14) are upper bounds; no asymptotic equivalence or finite trajectory length is asserted.

The ingredients have substantial precedent. Higher-degree reduced Lyapunov functions and stable-complement transfer are treated by Grushkovskaya [2, Theorem 3 and Remark 1]. Bartha and Garab [3, Theorem 4.4] prove attraction at an exact fourth-order resonance for another planar map with a strict cubic radial estimate. Conservative higher-order invariants are discussed by Gelfreich and Vieiro [4]. Our purpose is the explicit energy and actual-phase proof for (1); neither resonant attraction nor angular averaging is claimed as a new general principle.

2 The angular correction

Lemma 1. *The integrals in* (7) *satisfy* $0 < I_0 < 2$, $J_0 < -1/25$ *and* $c < -512/25$. *The function* η *is smooth and* $\pi/2$*-periodic.*

Proof. Put $\alpha = q/2$. Since

$$e \geq 1 - q/2 = \frac{1}{2+q} > \frac{1}{4},$$

all integrands are smooth and $0 < I_0 < 2$. Define

$$b(t) = 3t - 4t^2 + \alpha t^3.$$

Differentiation of a periodic function gives the exact identity

$$\frac{1}{4}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin(4\theta)t(\theta)e(\theta)^{-3/2}\right) = b(t(\theta))e(\theta)^{-5/2}. \tag{16}$$

Consequently $\langle be^{-5/2}\rangle = 0$. On $0 \leq t \leq 1$,

$$f + \frac{b}{8} = t\left(q - \frac{13}{8} + \frac{q-1}{4}t + \frac{q}{16}t^2\right) \leq -dt, \qquad d = \frac{30 - 21q}{16} > 0.$$

The bracket is increasing, and its value at one is $-d$; here $q < 10/7$. Thus

$$J_0 \leq -d\langle te^{-5/2}\rangle. \tag{17}$$

The binomial series for $(1-\alpha t)^{-5/2}$ converges uniformly on $[0,1]$ and has positive coefficients. Using its first four terms and $\langle t^j\rangle = \binom{2j}{j}/4^j$ yields

$$\langle te^{-5/2}\rangle > \frac{1}{2} + \frac{15q}{32} + \frac{175}{256} + \frac{3675q}{8192} = \frac{303}{256} + \frac{7515q}{8192} =: L_0.$$

Now

$$dL_0 = \frac{21834q - 24750}{131072} > \frac{1}{25},$$

because $q > 7/5$ and $25(21834 \cdot 7/5 - 24750) = 145440 > 131072$. This proves the stated bound for J_0; the bound for c follows from $I_0 < 2$.

The integrand defining η has period $\pi/2$ and its full-period integral is $2\pi(cI_0 - 1024J_0) = 0$. Each quarter-period integral is therefore zero. This proves the asserted periodicity and smoothness. ∎

Lemma 2. *The function B is branch independent and homogeneous of degree six. It is smooth away from zero and $C^{5,1}$ at zero. For some $\rho > 0$,*

$$\frac{\|z\|^4}{2(2+q)} \le U(z) \le 2\|z\|^4 \qquad (\|z\| < \rho). \tag{18}$$

Moreover, for a finite constant C and every z,

$$\|DU(z)\| \le C(\|z\|^3 + \|z\|^5). \tag{19}$$

Proof. Writing $z = r(\cos\theta, \sin\theta)$ gives

$$Q(z) = r^4 e(\theta), \qquad \frac{r^4}{2+q} \le Q(z) \le r^4. \tag{20}$$

Periodicity of η establishes branch independence. Every derivative of order j of B away from zero is homogeneous of degree $6 - j$, with bounded restriction to the unit circle. For $j \le 5$ these derivatives extend continuously to zero by zero. Inductively they are the actual derivatives there, since the difference quotients are bounded by positive powers of r. Fifth derivatives have bounded first derivatives off zero. Integrating along a segment, split at zero when necessary, proves that they are Lipschitz. No C^6 assertion is needed.

The angular factors of A and B are bounded, so $|A + B| \le C_6 r^6$ for a finite C_6. Combine this with (20) and choose ρ sufficiently small to obtain (18). Homogeneity of the first derivatives proves (19), also at zero. ∎

3 Actual finite iterates and their remainders

The potential identity (2) proves coercivity: bounded F bounds y and then $x + y^2$, hence x. Its critical equations give $x = -y^2$ and $y(1+2y^2) = 0$. The map itself is a global polynomial diffeomorphism. Indeed, from an output phase one recovers

$$x = x^+ - qv^+, \quad y = y^+ - qw^+, \quad v = (1+q)v^+ + q(x+y^2), \quad w = w^+ + q(y+2xy+4y^3).$$

Similarly Ξ is a polynomial diffeomorphism, with $y = 2X$, $w = q(X - Y)$ and $u = \sigma + H(z)$. Substitution gives exactly (5). The center linear block is the quarter-turn $R(X, Y) = (-Y, X)$. The trace, determinant and discriminant of S are $2 - q$, $q - 1$ and $10 - 8q < 0$; its two eigenvalues have modulus $\sqrt{q-1} < 1$.

Lemma 3. *Uniformly for sufficiently small $r = \|z\|$ and $s = \|\sigma\|$,*

$$\begin{aligned} \|z_4(z,\sigma) - z_4(z,0)\| &\le C_1 rs, \\ \|\sigma_4(z,\sigma) - S^4\sigma\| &\le C_2(r^2 s + r^6), \\ U(z_4(z,0)) - U(z) &= cQ(z)^2 + O(r^{10}), \end{aligned} \tag{21}$$

for finite constants C_1, C_2. These estimates concern the exact map, with no invariant-graph assumption.

Proof. At $\sigma = 0$ the exact first center update is $Rz + g_3(z) + g_5(z)$, where

$$g_3 = (-8(2+q)X^3 + 8qXY^2, 0), \qquad g_5 = ((256 - 128q)X^2Y(X^2+Y^2), 0).$$

The definitions of H_2, H_4 give

$$\begin{aligned} SH_2(z) + bX^2 - H_2(Rz) &= 0, \\ SH_4(z) - H_4(Rz) - DH_2(Rz)g_3(z) &= 0. \end{aligned} \tag{22}$$

For clarity, the first nonzero pure-center transverse residual is

$$\begin{aligned} R_{6,1} &= 256qX^6 + (2048q - 5120)X^4Y^2 + (3328q - 4096)X^2Y^4, \\ R_{6,2} &= (1024 + 256q)X^6 + (2048 - 5632q)X^4Y^2 + (7168 - 3328q)X^2Y^4. \end{aligned} \tag{23}$$

In particular $R_{6,1}(1,0) = 256q \neq 0$.

Here is a direct justification of uniform remainders. The center output is odd and the transverse output is even under $(z,\sigma) \mapsto (-z,\sigma)$. Also $\Psi(0,\sigma) = (0, S\sigma)$. Every term in $\sigma^+ - S\sigma$ has at least two z factors. Subtracting its value at $\sigma = 0$ leaves at least one σ factor as well. The cancellations (22) and finite polynomiality therefore give

$$\sigma^+ = S\sigma + R_*(z,\sigma), \qquad \|R_*(z,\sigma)\| \le C(r^2 s + r^6) \tag{24}$$

on a small product neighborhood.

For each fixed $j \le 4$, the polynomial z_j vanishes at $z = 0$, so $\|z_j\| \le C'r$ on a fixed compact product neighborhood. Shrink the input neighborhood so its first four images lie in the domain of (24). Induction then gives $\|\sigma_j\| \le C''(s + r^6)$. Substituting these bounds into

$$\sigma_4 - S^4\sigma = \sum_{i=0}^{3} S^{3-i} R_*(z_i, \sigma_i)$$

proves the second inequality of (21). The polynomial derivative $D_\sigma z_4$ vanishes at $z = 0$ and is $O(r)$ uniformly there. Integration along the parameter segment from zero to σ proves the first inequality.

For initial $\sigma = 0$, we have $\sigma_j = O(r^6)$. Its contribution $-4X_j\sigma_{j,1}$ to the center update is therefore of degree at least seven. Direct iteration through degree five yields

$$z_4(z,0) = z + G_3(z) + G_5(z) + O(r^7), \tag{25}$$

where

$$G_3 = \big(16qX^2Y - 16(2+q)Y^3,\ 16(2+q)X^3 - 16qXY^2\big) \tag{26}$$

and

$$\begin{aligned} G_{5,1} &= 384(1+q)X^5 - (4096+2048q)X^3Y^2 + (384+1152q)XY^4, \\ G_{5,2} &= 384(q-1)X^4Y - (1536+512q)X^2Y^3 + 128(1+q)Y^5. \end{aligned} \tag{27}$$

One way to verify these finite expressions is to set $a_0 = z$, $b_0 = d_0 = 0$ and successively use

$$a_{j+1} = Ra_j, \qquad b_{j+1} = Rb_j + g_3(a_j), \qquad d_{j+1} = Rd_j + Dg_3(a_j)b_j + g_5(a_j).$$

Then $a_4 = z$, $b_4 = G_3$, $d_4 = G_5$. The previously established actual transverse residual is retained in the $O(r^7)$ term. Odd parity excludes even center degrees, and the finite polynomial map makes the remainder uniform.

With $J(a,b) = (-b,a)$, direct differentiation gives

$$G_3 = 4(2+q)J\nabla Q, \qquad DQG_3 = 0, \tag{28}$$

and

$$\begin{aligned} E_8 &:= DQG_5 + \tfrac{1}{2}D^2Q[G_3,G_3] + DAG_3 \\ &= 4096(q-2)(X^6Y^2 + X^2Y^6) + 12288(q-1)X^4Y^4 \\ &= 1024r^8 f(\theta). \end{aligned} \tag{29}$$

The Hessian term in this formula is part of the actual discrete difference. Taylor expansion of the polynomials now gives

$$(Q+A)(z_4(z,0)) - (Q+A)(z) = E_8(z) + O(r^{10}).$$

To include B, observe from (26) that

$$D\theta\, G_3 = 16(2+q)r^2 e(\theta).$$

Equations (7) and (28) imply

$$DB\, G_3 = r^8\big(ce(\theta)^2 - 1024f(\theta)\big). \tag{30}$$

This computes a homogeneous Taylor coefficient. To justify it in the actual difference, put $\Delta = z_4(z,0) - z = G_3 + O(r^5)$. For small nonzero r, $\Delta = O(r^3)$ and the segment $z + t\Delta$ has norm comparable to r. Homogeneity gives $DB = O(r^5)$ and $D^2B = O(r^4)$ along that segment. Hence

$$B(z+\Delta) - B(z) = DB(z)G_3(z) + O(r^{10}).$$

Adding (29) and (30) proves the last assertion of (21). At $z = 0$ it is exact. In particular, no continuous averaged trajectory has been substituted for the discrete orbit. ∎

4 A strict energy in the original phase

Lemma 4. *The constant C_* in (10) is finite. The matrix P is positive definite and satisfies*

$$(S^4)^T P S^4 - P = -I. \tag{31}$$

Proof. Write $Z(z,\sigma) = z_4(z,\sigma)$. It is polynomial and $Z(0,\sigma) = 0$ for every σ. On the entire compact unit product, bounded polynomial derivatives therefore give

$$\|Z(z,\sigma)\| \le L_1 r, \qquad \|Z(z,\sigma) - Z(z,0)\| \le L_2 r s.$$

For the second estimate, $D_\sigma Z(0,\sigma) = 0$ and bounded mixed derivatives imply $D_\sigma Z = O(r)$, followed by integration in σ. By (19), the derivative of U on the segment between the two outputs is bounded by $C(r^3 + r^5)$. Since $r \le 1$, their energy difference is at most a finite constant times $r^4 s$. This proves finiteness of the stated supremum before K is used.

Direct multiplication verifies (31). For positive definiteness, its first diagonal entry is positive and

$$\det P = \frac{579 + 411q}{1088} > 0. \qquad \blacksquare$$

Proof of Theorem 1. The preceding lemmas give well posedness, regularity and $\kappa > 0$. Write $r = \|z\|$, $s = \|\sigma\|$. By (20) and Lemma 3, shrinking the neighborhood gives

$$U(z_4(z,0)) - U(z) \le -\frac{\kappa}{2} r^8.$$

Write $\sigma_4 = S^4\sigma + R$ with $\|R\| \le C_2(r^2 s + r^6)$. The exact identity (31) implies

$$\sigma_4^T P \sigma_4 - \sigma^T P \sigma = -s^2 + 2(S^4\sigma)^T P R + R^T P R.$$

The additional terms are bounded by constant multiples of $r^2 s^2 + r^6 s + r^4 s^2 + r^{12}$. Young's inequality first absorbs $r^6 s$ with an arbitrarily small multiple of s^2 and a finite multiple of r^{12}. A sufficiently small r then absorbs the $r^2 s^2$ and $r^4 s^2$ terms. Thus, for a finite C_3,

$$\sigma_4^T P \sigma_4 - \sigma^T P \sigma \le -\frac{3}{4} s^2 + C_3 r^{12}. \tag{32}$$

The definition of C_* and Young's inequality give, for $r, s \le 1$,

$$|U(z_4(z,\sigma)) - U(z_4(z,0))| \le C_* r^4 s \le \frac{\kappa}{8} r^8 + \frac{2C_*^2}{\kappa} s^2 \le \frac{\kappa}{8} r^8 + \frac{K}{4} s^2.$$

Consequently

$$W_0(\Psi^4(z,\sigma)) - W_0(z,\sigma) \le -\frac{3\kappa}{8} r^8 - \frac{K}{2} s^2 + K C_3 r^{12}.$$

Here K is already a fixed finite number. Reduce the radius so that $K C_3 r^4 \le \kappa/8$. We obtain

$$W_0 \circ \Psi^4 - W_0 \le -\frac{\kappa}{4} r^8 - \frac{K}{2} s^2. \tag{33}$$

By (18) and positivity of P, W_0 is locally comparable to $r^4 + s^2$. Choose the input neighborhood so its first three images are in that positivity neighborhood. The term $j = 0$ gives the lower bound for Φ. The finite-iterate estimates $\|z_j\| \le Cr$ and $\|\sigma_j\| \le C'(s+r^6)$ give its upper bound. Thus constants $m, M > 0$ yield the first line of (12). The identity

$$\Phi \circ \Psi - \Phi = W_0 \circ \Psi^4 - W_0$$

is exact, and (33) gives the second line with $a = \min\{\kappa/4, K/2\}$. This elementary finite-sum conversion is the reason for the four terms in (11).

We next prove capture before iterating this local inequality. Choose $R_0 < 1$ so the closed ball of radius R_0 is inside the valid neighborhood. Continuity and $\Psi(0) = 0$ give $0 < \rho < R_0$ with $\Psi(B_\rho) \subset B_{R_0}$. The number

$$\beta = m \min\{\|z\|^4 + \|\sigma\|^2 : \rho \le \|(z,\sigma)\| \le R_0\}$$

is positive. Define

$$\Omega = \{(z,\sigma) \in B_\rho : \Phi(z,\sigma) < \beta/2\}.$$

It is an open neighborhood of zero. For an input in Ω, its output lies in B_{R_0} by the finite no-jump condition. The local decrement puts its energy below $\beta/2$; the annular lower bound excludes an output outside B_ρ. Thus $\Psi(\Omega) \subset \Omega$, and $\mathcal{B} = \Xi^{-1}(\Omega)$ is invariant under the original map. This argument applies in arbitrarily small valid R_0-balls, proving original-phase Lyapunov stability.

Only after capture do we set $E_n = \Phi(\Psi^n(\Xi(p_0)))$. Since $s_n \le 1$, the upper comparison gives

$$E_n^2 \le 2M^2(r_n^8 + s_n^4) \le 2M^2(r_n^8 + s_n^2).$$

Therefore $0 \le E_{n+1} \le E_n - kE_n^2$, where $k = a/(2M^2) > 0$. While the energy is nonzero,

$$\frac{1}{E_{n+1}} - \frac{1}{E_n} = \frac{E_n - E_{n+1}}{E_n E_{n+1}} \ge k.$$

If the energy reaches zero, local positive definiteness forces the zero state, which is fixed. These two cases give (13) and convergence to zero. They also give $r_n = O(n^{-1/4})$ and $s_n = O(n^{-1/2})$. Finally

$$(y_n, w_n) = (2X_n, q(X_n - Y_n)), \qquad (x_n, v_n) = \sigma_n + H_2(z_n) + H_4(z_n)$$

prove (14). All zero-coordinate cases are included in the preceding inequalities. ∎

5 The polynomial-correction obstruction

Proof of Theorem 2. Let $P_6 = Q_6 - A$ and define $\mathcal{L}P = DPG_3$. The actual expansion (25), the cancellation $DQG_3 = 0$ and (29) show that the limit exists and is

$$D_{Q_6} = E_8 + \mathcal{L}P_6. \tag{34}$$

It is a homogeneous polynomial of degree eight. At $z = 0$ the limit is zero; the expansion also applies as ε tends to zero through negative values.

To force a negative value, consider the compact smooth oval $Q = 1$. The polynomial field $G_3 = 4(2 + q)J\nabla Q$ is tangent to it. Its angular velocity there is $16(2 + q)e^{1/2} > 0$, so it traverses the oval periodically in finite time. The integral of $\mathcal{L}P_6$ around this closed orbit is zero. Since $r = e^{-1/4}$ on the oval,

$$\oint D_{Q_6}\,\mathrm{d}\tau = \frac{64}{2+q}\int_0^{2\pi} f e^{-5/2}\,\mathrm{d}\theta = \frac{128\pi}{2+q}J_0 < 0. \tag{35}$$

Here τ is the auxiliary field's time, used only to integrate a polynomial coefficient. Equation (35) therefore implies a strictly negative value of D_{Q_6}; it makes no assertion about a continuous approximation to an original orbit.

For a positive value one may use a finite exact identity. Write $Q_6 = \sum_{j=0}^{6} a_j X^{6-j}Y^j$. Substitution in (34) gives

$$\begin{aligned} D_{Q_6}(1,0) &= 16(2+q)a_1 + 512(1+q), \\ D_{Q_6}(0,1) &= -16(2+q)a_5 - 512(1+q), \\ D_{Q_6}(1,1) + D_{Q_6}(1,-1) &= 256(a_5 - a_1) + 8192(5q-7). \end{aligned}$$

Hence

$$\frac{16}{2+q}\big(D_{Q_6}(1,0) + D_{Q_6}(0,1)\big) + D_{Q_6}(1,1) + D_{Q_6}(1,-1) = 8192(5q-7) > 0.$$

All weights are positive and $5q > 7$, so one of these four values is positive. Together with (35), this proves precisely the sign change asserted in the theorem. ∎

6 Scope and source comparison

The conclusion is local for the single fixed recurrence (1). The potential's global coercivity and the global polynomial inverse do not give a global attracting basin. The proof gives an open captured neighborhood through positive local energy and a finite no-jump condition. No maximal radius or numerical value for C_* is asserted. The angular term's regularity suffices for the exact Taylor estimates; the result does not prove that higher regularity is impossible for a different energy.

Classical analysis uses the fourth iterate and retains the resonant monomials [1, 5]. Grushkovskaya's Theorem 3 and Remark 1 [2] already give conditional higher-order reduced-to-full Lyapunov transfer before the additional nonresonance restriction used for a later radial model. The proof here writes the transfer directly for a finite polynomial graph, retaining its nonzero defect. Bartha–Garab's fourth-order resonant example [3] supplies a genuine attraction precedent; its strict cubic radial estimate differs from the quartic-level calculation used here. The conservative analytic setting of [4], including its

degenerate resonance example and formal invariant construction, supplies related energy ancestry rather than a damping conclusion for this map.

The inspected sources do not provide an identified theorem for the particular energy and fixed recurrence above. That bounded comparison is not a priority certificate. The complete Skeel–Srinivas paper [6] and Neishtadt paper [7], which are close classical comparators for numerical or fourth-order resonances, were not available for theorem-level comparison in this study. Their titles and bibliographic records are therefore not used as evidence that they exclude or contain the present result. The accompanying verification guide specifies the inspected source versions and reading scope. We make no firstness claim or general resonance classification.

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